

Apparel Product Tech Pack

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Materials, construction, measurements, lay planning and costing coursework.

Academic research and concept work; not a commissioned commercial engagement.

The original student-ID cover has been replaced. Source references are retained.

Recommendations and estimates are not verified business outcomes.

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01 Introduction

Apparel brands need to standardize their mass-produced garments, so this optimally designed, mass-produced garment production technology package is developed and used to give manufacturers a standardized guide. With this technical package description, manufacturers can know exactly what fabrics are needed for the garment, how the fabric is utilized and cut, trim, cost, sizing, sewing or seams, etc. (LaBat, 2007) The technical package details how manufacturers should standardize production to avoid some production errors, thus ensuring garment quality and output time, and avoiding the resulting cost risks. If the manufacturer deviates from the production of the technical package, the brand can also impose some penalties on the manufacturer's deviation, or even withdraw the order, according to the standardization criteria of the production package. ZARA, a Spanish clothing brand, is a high fashion brand with about 650 stores in 50 countries and regions. As a high fashion fast fashion brand, ZARA needs to achieve high fashion with appropriate pricing and respond quickly to market replenishment. This sales concept requires manufacturers to control production costs while achieving the highest quality, according to a technical package, and to be able to quickly replenish stock in volume. (Ferdows, Lewis and Machuca, 2003) As ZARA, whose target customers are mainly young women, shorts have been its best-selling items. zara linen shorts are oriented to young consumers while also deriving the age group upwards, with loose and breathable characteristics in summer, and using linen and cotton combination or linen and viscose combination in fabric or lining, etc., with more characteristics in fabric texture, so that consumers have more choices. (Ferdows, Lewis and Machuca, 2005) Figures 1, 2 and 3 are the three yama shorts in Zara and their comparison in terms of price, 2 color, fabric, etc.

The shorts in this technopack are very similar to Figure 2, with some seasonal changes in fabric cut, decoration and other details to correspond to fashion trends. For example, the shorts in this technoboard make multiple pleats on the seam allowance, 2 on the front and 4 on the back. The cost of these extra pleats is limited and can be satisfied by using only more yarn, so the cost can be controlled; moreover, it can improve the breathability of shorts in spring and summer, so that consumers can wear them more comfortably and breathably; more pleats will make the shape more A-type loose and fashionable, following the fashion trend. The retail price of the linen shorts of this technical package is 35 pounds, the fabric is composed of 52% linen, 48% cotton, the size range is xxs-xxl, its washing requirements are maximum machine wash 30°C/86°F, short spin cycle, do not use bleach, iron at maximum 110°C/230°F, dry clean with perchloroethylene not tumble dry. The fabric layout chart of this technical package is well arranged, and the reasonable arrangement allows manufacturers to reduce fabric loss. For example, the layout of cotton and linen fabric and the use of cutting in this technical package can both make the fabric utilization rate up to 80% and also point out the way of cutting fabric to reduce the damage to cotton and linen fabric, thus controlling the quality of shorts. Moreover, the stitching and seams in it are clear, which can make manufacturers understand more about the detailed production process and make the shorts more detailed, so that the quality of the shorts will be better and the fastest production can be achieved.



High-waisted Bermuda shorts: made of linen blend. Side pockets and patch false pockets on the back, metal hook and loop, buttons and zippers on the front.

Size: xxs-xxl
Fabric: 50% viscose – 50% linen
Preservation: Do not use bleach, iron at max 110°C/230°F, dry clean with perchloroethylene, do not tumble dry.
Price: £59.99



High-waisted bermuda shorts made of linen blend. Side pockets and false patch back pockets. Darts detail. Front zipper closure with inside button and metal hook and loop closure.

Size: xxs-xxl
Fabric: 52% linen – 33% cotton – 15%
Care: Machine wash 30°C/86°F max, short spin cycle, do not use bleach, iron at 110°C/230°F max, dry clean with perchloroethylene, do not tumble dry
Price: £29.99



Linen blend with waistband Bermuda shorts

Size: xxs-xxl
Fabric: at least 50% European grown linen and 50% Join Life viscose fiber.
Care: Machine wash at a maximum of 30°C/86°F, short spin cycle, do not use bleach, iron at a maximum of 110°C/230°F, dry clean with perchloroethylene, do not tumble dry
Price: £25.99

02 WORKING DRAWING

SAMPLE GARMENT: Basic Shorts

DEPARTMENT: LADIES SHORTS

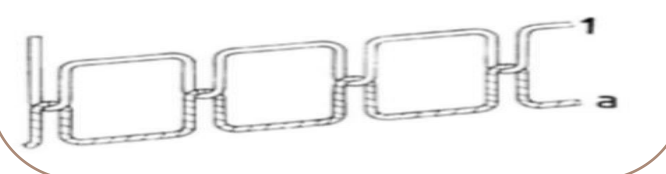
DATE/SEASON: SS 23/24

Front

Zip, use invisible nylon zipper, length 19.5cm

Lapped Zip, using 301 lockstitch(also for top stitching)

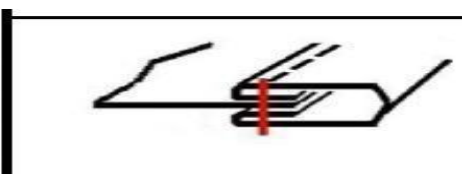
301



Two Tucks, using 301 lockstitch

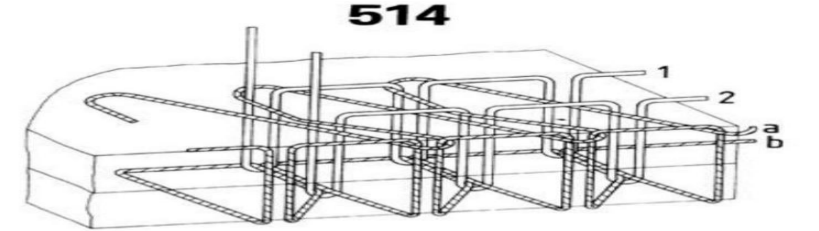
Separate Waistband, using 301 lockstitch

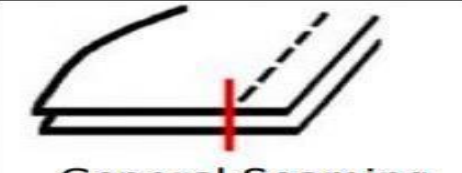
Attach the waistband by using 301 lockstitch (with BSc–3.05.01 seaming)

	BSc	3.05.01
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Side seams, first use 514 overlook, then use 301 lockstitch sewn together (using general seaming–SSa 1.01)

514



	SSa	1.01
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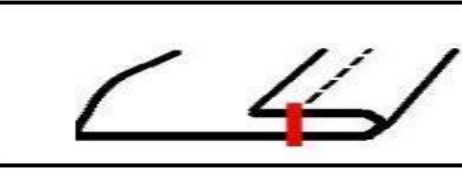
General Seaming

Back

Four Back Darts, using 301 lockstitch

Centre back, first use 514 overlook, then use 301 lockstitch (using general seaming–SSa 1.01)

Lock stitch & Overlock hem, first use 514 overlook, then use 301 lockstitch (using EFa–6.02.01 seaming)

	EFa	6.02.01
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Herm Turn up

Bar Tack at Hem Turn up, use 301 lockstitch (using EFa–6.02.01)

Any other information:

301 stitch is the main stitch of shorts, it is very widely used in garment production and is a cost-effective option for splicing two pieces of fabric together. (Rasheed et al., 2014) It can ensure that the upper and lower two pieces of cloth have a certain sewing strength at the same time, saving the manufacturer's sewing material.

514 stitch is commonly used in seams, the stitch is connected by an uninterrupted line and a ring of lines, its link effect is better, suitable for highly ductile materials. (Niroshan, 2021) Shorts material is cotton and linen, 514 seams can be effectively spliced cotton and linen these ductile materials, to prevent the extension process caused by the off-line, while the pattern is better looking. The 514 seams can effectively splice cotton and linen, which are malleable materials, to prevent stripping caused by the stretching process, while the pattern is better looking.

The 3.50.01 seam type is a very strong and durable seam type that is often used for piping, and is suitable for splicing shorts and waistbands. (Webster, Laing and Niven, 1998)

1.01 seam type is very common and is the main seam in shorts, which can be used to stack two pieces of cloth together with a very low amount of thread. (Barbulov-Popov et al., 2012)

6.02.01 seam type is a common hemming seam type, which can be customized. The width of the wrap hem, which is elaborated in terms of aesthetics, is suitable for Zara's positioning.

03 LAY PLAN

SAMPLE GARMENT STYLE: Basic Shorts

DATE/SEASON: SS23/24

GARMENT SIZE: BUSINESS:

NO	PATTERN NAME	TOTAL NUMBER OF PATTERN PIECES	INSTRUCTIONS to CUTTER
1	Front	Cut1	Cut on the straight grain
2	Waistband	Cut1 Calico & 1 Interfacing	Cut on the straight grain
3	Back	Cut2	Cut on the straight grain
4	Zipgaurd	Cut1	Cut on the straight grain



Any other information:

Shorts material is composed of linen and cotton together, a single fabric clothing fabric utilization rate is higher, and linen fabric is generally first extracted from linen fibers, these fibers make the adhesion of fiber bundles to get part of the loose, and then by pressing and rolling, playing hemp processing into "playing into the hemp", "playing into the hemp" for 10 to 20 single fibers composed of process fibers.(Buschle-Diller et al., 1994) The fabric cost is higher, and the fabric utilization rate of the shorts is relatively low for both combinations of fabrics. Therefore, it is important to have a reasonable layout of the fabric plan. When manufacturing in bulk, the fabric should be stacked in layers for layout, and the most reasonable layout can save more fabric.(Syduzzaman and Golder, 2015) Using machine electric cutting, fabric utilization layout will be more efficient, fabric orientation and angle can also increase the utilization rate, according to the characteristics of cotton and linen in accordance with a certain angle of the same

orientation cutting.

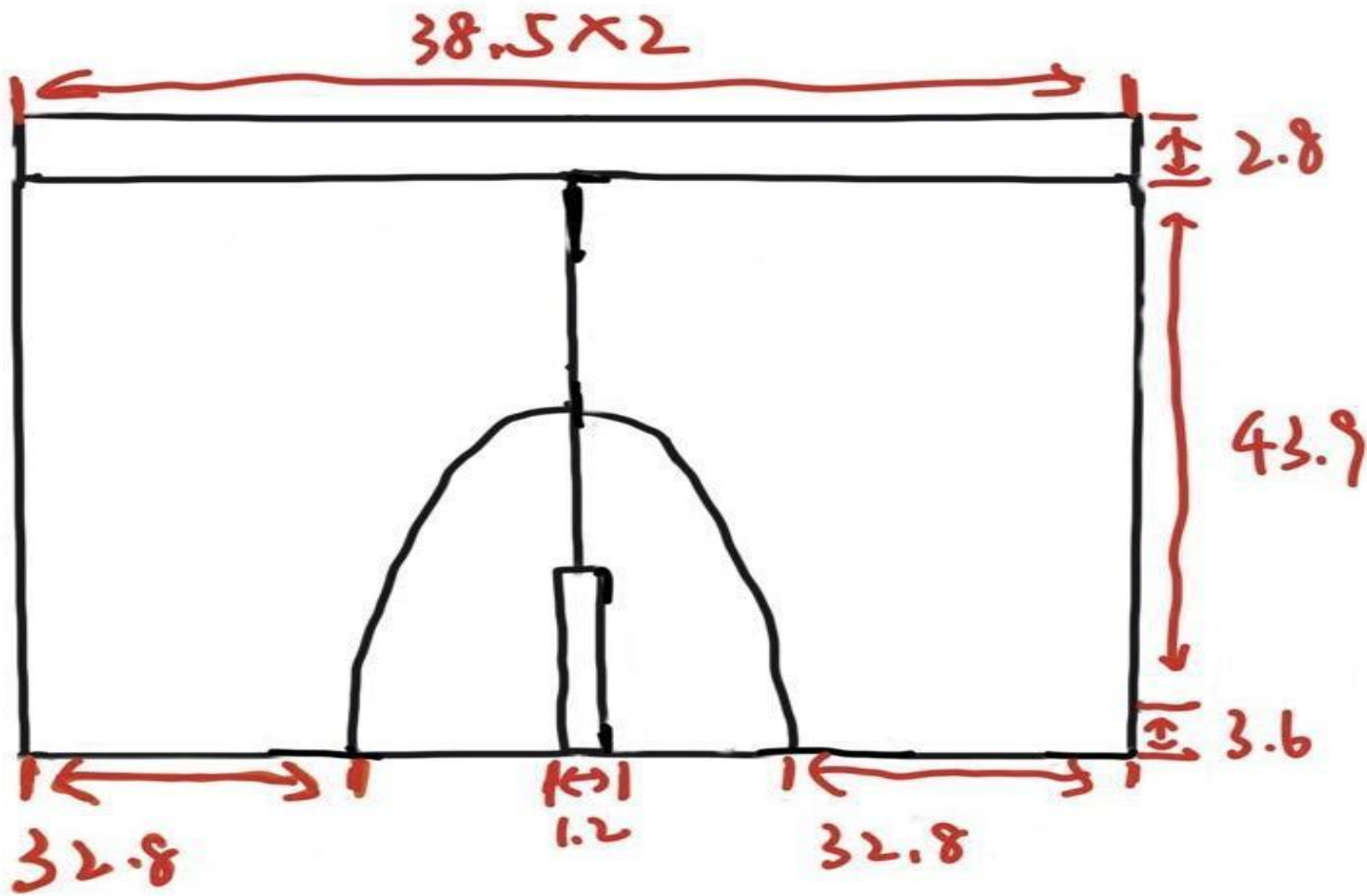
FABRIC TYPE: a
linen and cotton
blend

FABRIC
COMPOSITION:52%
linen and 48% cotton

WIDTH OF FABRIC
CM:50.3cm

TOTAL FABRIC
LENGTH CM USED
=77cm

SAMPLE LAY PLAN



Size of material used (full scale)
Length= $38.5\text{cm}^2=0.77\text{m}$ (full scale)
Width= $2.8\text{cm}+43.9\text{cm}+3.6\text{cm}=0.503\text{m}$ (full scale)

04 Costings

SAMPLE GARMENT: Basic Shorts

DEPARTMENT: LADIES SHORTS

DATE/SEASON: SS 23/24

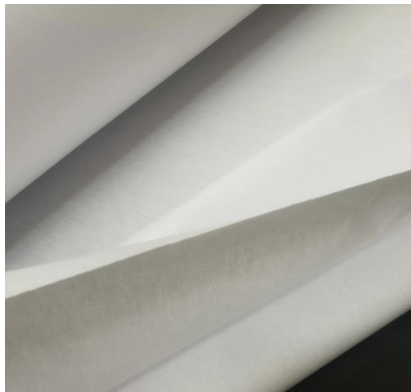
FABRIC/TRIM/ATTACHMENTS	COMPOSITION	AMMOUNT USED in Metre	COST per Item/ Metre	SUPPLIER
A linen and cotton blend	52% linen and 48% cotton	0.77m(width 0.503m)	£7.69	WeLoveFabricUK https://www.etsy.com/uk/listing/1001613170/premium-natural-plain-cotton-linen-blend?ga_order=most_relevant&ga_search_type=all&ga_view_type=gallery&ga_search_query=linen+cotton+blend+fabric&ref=sr_gallery-1-1&frs=1&bes=1&organic_search_click=1
INTERFACING/FUSIBLE	WASHABLE IRON ON	0.7m(width 0.43m)	£0.315	Vilene
ZIP	PLASTIC SPIRAL TEETH GAUGE 12	1	£0.5	HaberdasheryCart https://www.etsy.com/uk/listing/563246152/2-hole-natural-wooden-buttons-x-10-pack?ga_order=most_relevant&ga_search_type=all&ga_view_type=gallery&ga_search_query=button&ref=sr_gallery-1-8&frs=1&edd=1&pop=1&organic_search_click=1
THREAD YARN	100% MULTI FILAMENT	156m	£0.1716	A&E
		Total/Garment	£8.6766	

Sample Materials:

6



A linen and cotton blend



Fusible interlining



Zip



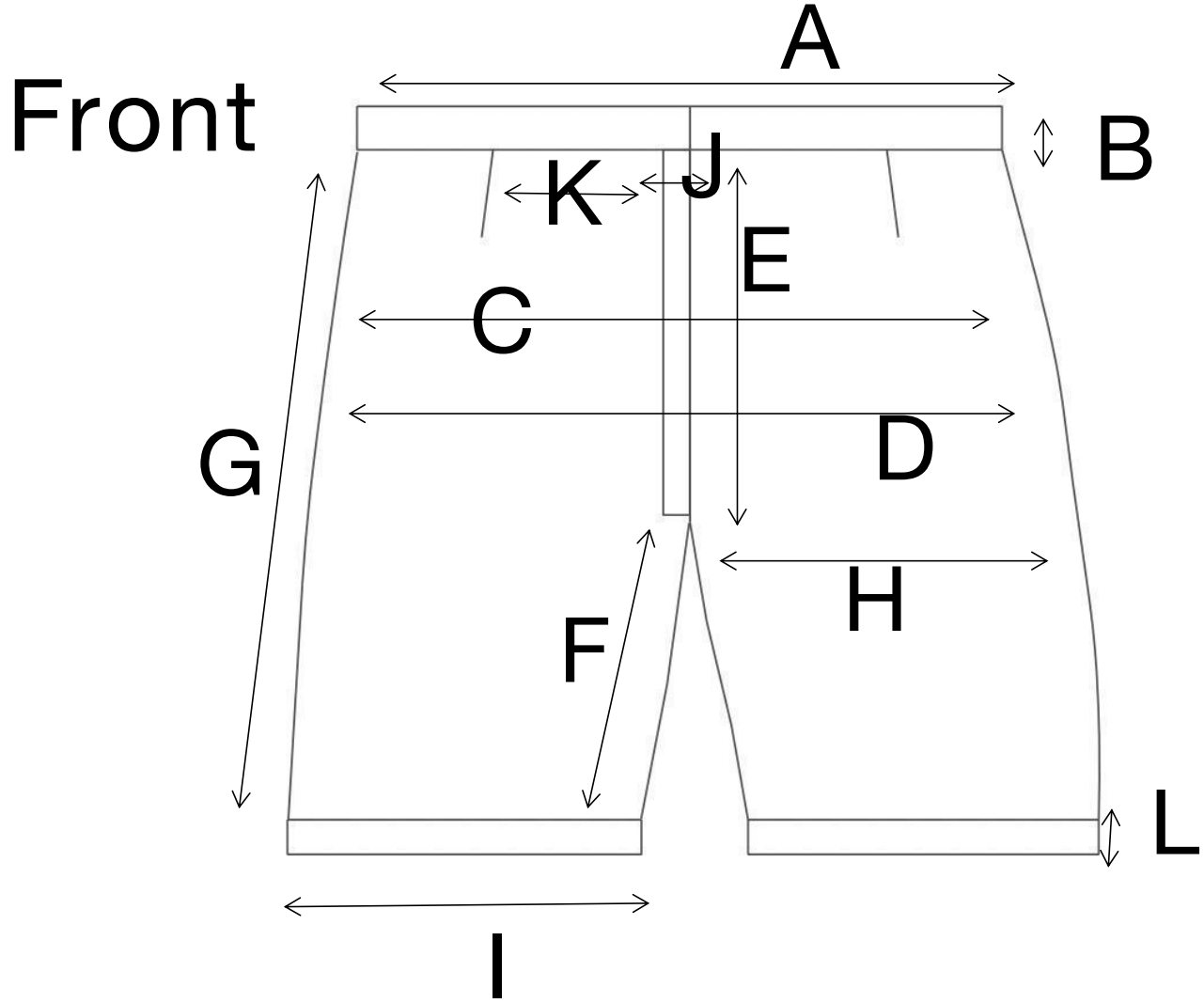
Thread yarn

Any other information:

The costing on the list does not include logistics costs as well as communication costs The cost price of the sample should consider the scale efficiency of the market, once the market is too large, the sample cost is too high to automatically eliminate sub-samples. The utilization rate can be improved through a reasonable layout of fabric flatness, and improved utilization can save money.(Singh and Nijhar, 2015)

05 Size Specification

SAMPLE GARMENT: Basic Shorts	
DEPARTMENT: LADIES SHORTS	DATE/SEASON: SS 23/24



LANDMAR K	MEASURMENT DESCRIPTION	CM	TOLERANCES
	FRONT		
A	WAISTLINE SEAM CIRCUMFERENCE(Reiaxed/Extended)	37.6/38.5	+/-1
B	WAISTBAND-HEIGHT	2.8-3.0	+/-0
C	HIGH HIP(4 in,or 10.16 cm,from natural waist)	47.8	+/-1
D	LOW HIP(7in.or 17.78 cm,from natural waist)	52.3	+/-1
E	RISE/ZIPPER PLA Longth	35.1/19.5	+/-1
F	INSEAM	11.9	+/-1
G	OUTSEAM	43.9	+/-1
H	THIGH CIRCUMFERENCE	33.7	+/-1
I	ANKLE CIRCUMFERENCE	32.8	+/-1
J	ZIPPER PLACKEI Width	1.2	+/-0
K	PLEAT Placement -Distance from CF/Pleat depth	10.4/5.6	+/-1
L	CUFF-HEIGHT/HEM-DEPTH	3.6/5.9	+/-0

05 Size Specification

SAMPLE GARMENT: Basic Shorts

DEPARTMENT: LADIES SHORTS

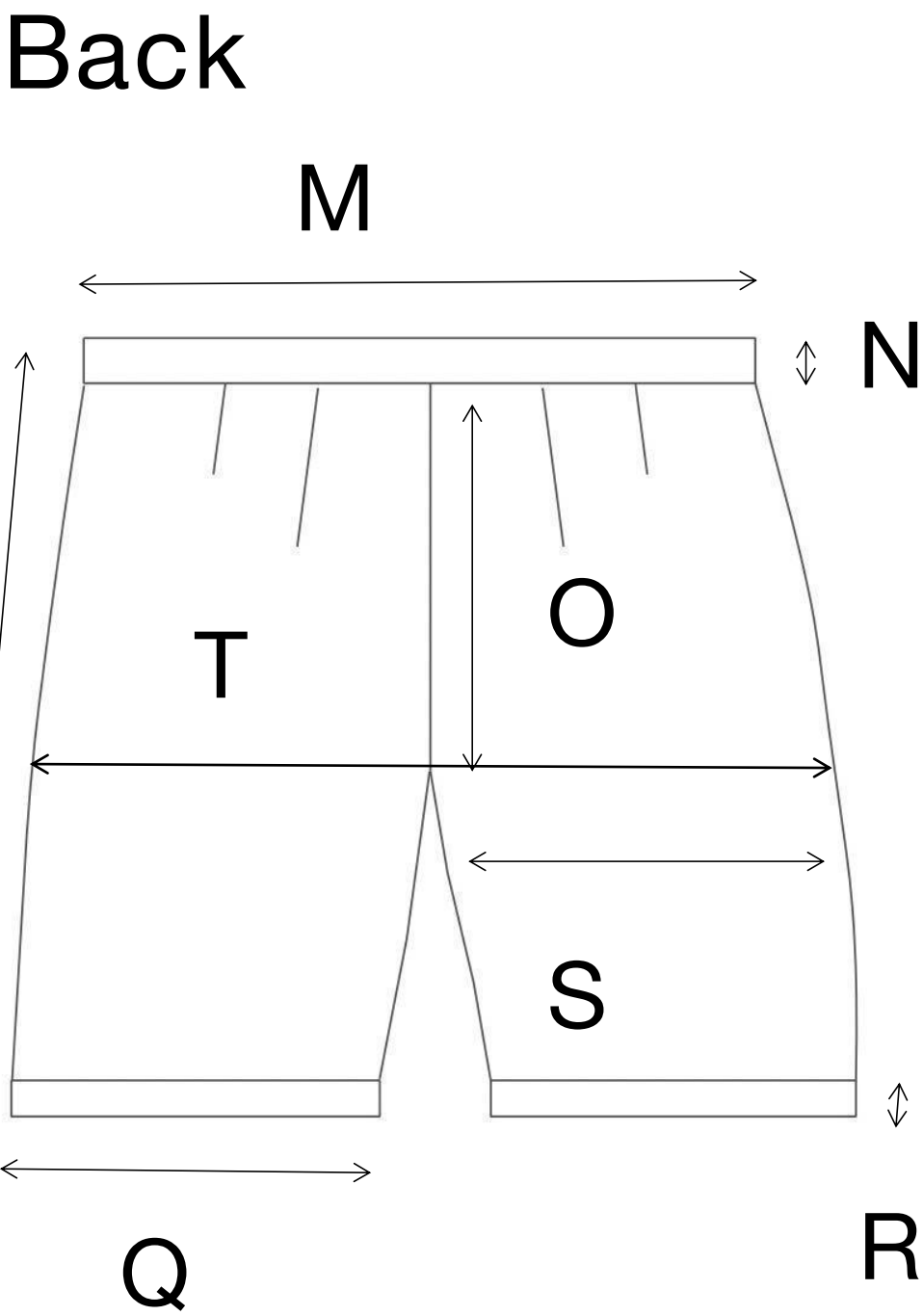
DATE/SEASON: SS 23/24

ANY OTHER INFORMATION

The direct error in sizing is very important and the tolerance of each size between xxs-xxl must be accurate. Here is the zara size chart showing the garment specification guide.(Ghemawat, Nueno and Dailey, 2003)

CLOTHING							
Size	6	8	10	12	14	16	18
US Size	26	27	28	29	30	31	32
European Size	34	36	38	40	42	44	46
Chest (inches)	32.3	33.9	35.4	37	38.6	40.2	41.7
Waist (inches)	25.2	26	27.6	29.1	30.7	32.3	33.9
Hips (inches)	35.4	37	38.6	40.2	41.7	43.3	44.9

Size	XXS	XS	S	M	L	XL	XXL
Chest (inches)	31.5	32.3	33.9	35.4	37.8	40.2	42.5
Waist (inches)	22.8	24.4	26	27.6	29.9	32.3	34.6
Hips (inches)	33.9	35.4	37	38.6	40.9	43.3	45.7



LANDMAR K	MEASURMENT DESCRIPTION	CM	TOLERANCES
	BACK		
M	WAISTLINE SEAM CIRCUMFERENCE(Reiaxed/Extended)	37.6/38.5	+/-1
N	WAISTBAND-HEIGHT	2.8-3.0	+/-0
O	RISE	35.1	+/-0
P	OUTSEAM	43.9	+/-1
Q	ANKLE CIRCUMFERENCE	32.8	+/-1
R	CUFF-HEIGHT/HEM-DEPTH	3.6/5.9	+/-0
S	THIGH CIRCUMFERENCE	33.7	+/-1
T	LOW HIP(7in.or 17.78 cm,from natural waist)	52.3	+/-1

06 Garment Breakdown

SAMPLE GARMENT: Basic Shorts

DEPARTMENT: LADIES SHORTS

DATE/SEASON: SS 23/24

Op No	Operation – Description	Machine type	Seam Classification	Thread Type	I.S.O Stitch Type	Needle Type & Size	S.P.IStitches per Inch	Attachment	Tolerance	S.ASeam Allowance
1	Pleat stitch the front and back of the shorts. Identify the cut and point markings on the fabric, fold the two cuts in half, sew from the cut to the point to form a pleat, and sew. (2 passes on the front and 4 passes on the back, following where the cut and point marks are)	Lockstitch	EFa 6.02.01	Corespun 120s	301 lockstitch	Sharp 0/8-90/14	10-12	-	+/-0	-
2	The two edges used for the center of the front and back, wrapped edge processing	4 Thread Overlock	EFa 6.02.01	Corespun 120s	514 4thread overlock	Sharp 0/8-90/14	10-12	-	+/-1	-
3	Match the two pieces of fabric together near the zipper, and sew the zipper part down at the waist and above the notch	Lockstitch	Plain seam ISO 01.01.01	Corespun 120s	301 lockstitch	Sharp 0/8-90/14	10-12	-	+/-1	1.5cm on both sides
4	Press open the centre back seam	Iron Press	-	Coresupn 120s	-	-	-	-	+/-1	-

06 Garment Breakdown

SAMPLE GARMENT: Basic Shorts

DEPARTMENT: LADIES SHORTS

DATE/SEASON: SS 23/24

Op No	Operation – Description	Machine type	Seam Classification	Thread Type	I.S.O Stitch Type	Needle Type & Size	S.P.IStitches per Inch	Attachment	Tolerance	S.ASeamAllowance
5	Perform zipper sewing on shorts. First of all, put the zipper aligned in the correct position, you can fix the zipper with the help of a needle on top and sew it. At the halfway point, you can pull the zipper open without destroying the straight stitching. After sewing, then carry out 90 degree reinforcement sewing at the bottom of the zipper.	Lockstitch	Plain seam ISO 01.01.01	Corespun 120s	301 lockstitch	Sharp 0/8-90/14	10-12	Zipper	+/-1	-
6	Fold the the zip guards width in half and overlock around the edge	Overlock	-	Corespun 120s	514 4thread overlock	Sharp 0/8-90/14	10-12	-	+/-1	-
7	Stitch the zip guard on the right side of the zip tape only	Lockstitch	Plain seam ISO 01.01.01	Corespun 120s	301 lockstitch	Sharp 0/8-90/14	10-12	-	+/-1	-
8	Se_x0008_w the seams on the front and back and press them apart	Iron press and lockstitch	Plain seam ISO 01.01.01	Corespun 120s	301 lockstitch	Sharp 0/8-90/14	10-12	-	+/-1	1cm

06 Garment Breakdown

SAMPLE GARMENT: Basic Shorts

DEPARTMENT: LADIES SHORTS

DATE/SEASON: SS 23/24

ANY OTHER INFORMATION

A y-shaped stitch can make the fabric stronger and reduce the waste caused by stitching.

Tolerance is the difference between the minimum and maximum allowed values of a specification, and the tolerance of a size can also make the difference between sizes smaller, allowing for a better fit of the shorts.(Barbulov–Popov et al., 2012)

Op No	Operation – Description	Machine type	Seam Classification	Thread Type	I.S.O Stitch Type	Needle Type & Size	S.P.IStitches per Inch	Attachment	Tolerance	S.ASeamAllowance
9	Locked hem on the bottom of the trouser leg	Overlock	-	Cores upn 120s	514 4thread overlock	Sharp 0/8-90/14	10-12	-	+/-1	-
10	Turn up the edge of the trouser leg, sewing processing	Lockstitch	EFa– 6.02.01 seaming	Cores upn 120s	301 lockstitch	Sharp 0/8-90/14	10-12	-	+/-1	1cm
11	Fold the pant legs over and at the seam edges, lock the folds	Lockstitch	EFa– 6.02.01 seaming	Cores upn 120s	301 lockstitch	Sharp 0/8-90/14	10-12	-	+/-1	2cm

07 Reference

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THANK YOU!

